

SAMPLE IDENTIFICATION	
Code	Faro 41
Well	2-LFST-1-AM
Depth	2.194,00 m
Coordinates (well)	Lat. 1°57'20,02" S Long. 56°58'17,00" W
Formation / Basin	Faro / Bacia do Amazonas
Age	Tournaisian–Visean (Lower Carboniferous)

GENERAL DESCRIPTION OF THE BLADE
More homogeneous and better selected sample. Grains with notable regularity in size and shape, clear contours, good individualization. Scarce intergranular material, in which direct contacts between grains with discrete brownish skins predominate. There are no differential concentrations of matrix, fractures or banding. The dark background between grains corresponds to grains in extinction position (Qz) and subordinately opaque/interstices. Lower proportion of ferruginous cement.

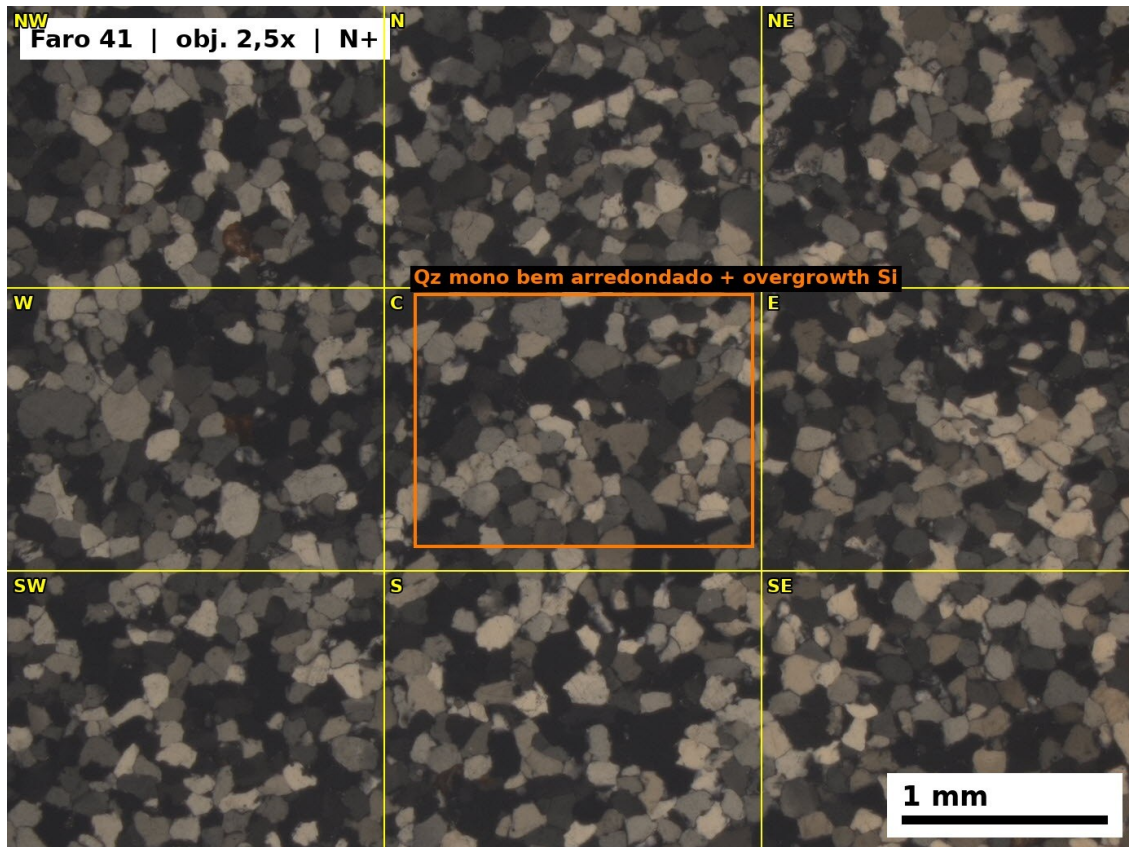
ESTIMATED MODAL COMPOSITION (% vol)	
Single-crystal quartz	~75–80%
Polycrystalline quartz	~3–5%
Feldspato (K-feldspato + plagioclásio)	~3–5%
Lithic fragments	~2–3%
Micas (muscovite ± biotite)	~1–2%
Opaque minerals	~2–3%
Cement (siliceous > ferruginous)	~5–7%
Clayey matrix	<2%
Visible porosity	<1%

Composition:
Predominantly quartzose (~75–80% monocrystalline quartz), with a minor fraction of unstable grains (feldspar + lithics <10%): High maturity consistent with facies representing maximum reworking (shoreface/foreshore or coastal bars reworked by storm waves). Dominant siliceous cementation, typical of clean quartzose sandstones.

MICROSCOPIC DESCRIPTION	
Texture	Clástica terrígena, grão-suportado, textura homogênea, excelente individualização.
Particle size distribution	Medium sand (~0.20–0.45 mm).
Particle size selection	Moderately well-selected.
Packaging and contacts	Dense. Long (~50%), concavo-convex (~30%), point (~15%), and sutured (~5%) contacts. Consistent with advanced burial (~2,194 m). Long contacts dominant in clean quartzose framework.
Cement / diagenesis	Dominant siliceous cementation (quartz overgrowths; juxtaposition of quartz grains with diffuse boundaries).

	Subordinate ferruginous cementation (isolated brownish spots). The dominance of silica reflects: a clean quartz framework supplying silica via pressure-solution; and lower residual permeability limiting the circulation of Fe-rich fluids.
Porosity	No evident open primary porosity. Residual porosity greatly reduced by compaction and siliceous cementation.
Maturity	Textural: mature to supermature (good sorting, subrounded to rounded, matrix <2%). Compositional: supermature (>75% Qz). The highest maturity among the samples.

ANÁLISE SETORIAL DE GRÃOS (grade 3×3)					
Setor	Mineral	Tam. (mm)	Esfericidade	Arredondamento	Forma
NW	Qz mono	0,30	High	Sub-rounded	Equidimensional
NW	Qz mono	0,25	High	Rounded	Equidimensional
N	Qz mono	0,35	High	Rounded	Equidimensional
N	Qz mono	0,22	Medium	Sub-rounded	Equidimensional
NE	Qz mono	0,30	High	Sub-rounded	Equidimensional
NE	Feldspato	0,25	Medium	Subangular	Prismatic
W	Qz mono	0,35	High	Rounded	Equidimensional
W	Qz mono	0,20	Medium	Sub-rounded	Equidimensional
C	Qz mono	0,30	High	Rounded	Equidimensional
C	Qz mono	0,28	High	Sub-rounded	Equidimensional
C	Opaco	0,10	High	Rounded	Equidimensional
E	Qz mono	0,35	High	Rounded	Equidimensional
E	Qz poli	0,25	Medium	Subangular	Elongated
SW	Qz mono	0,30	High	Sub-rounded	Equidimensional
SW	Qz mono	0,25	High	Rounded	Equidimensional
S	Qz mono	0,28	High	Sub-rounded	Equidimensional
S	Lítico	0,30	Low	Subangular	Elongated
SE	Qz mono	0,32	High	Rounded	Equidimensional
SE	Qz mono	0,22	Medium	Sub-rounded	Equidimensional
SE	Qz mono	0,35	High	Rounded	Equidimensional



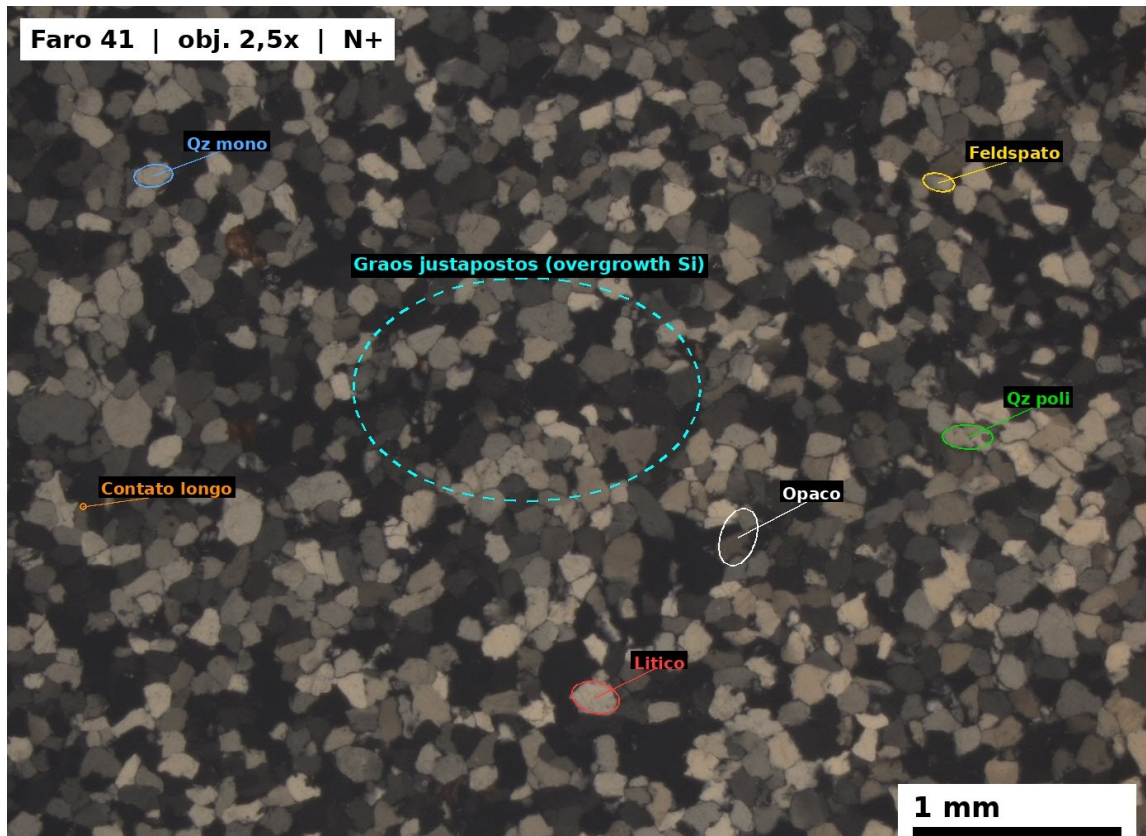
Grain population:

A practically unimodal population of monocrystalline quartz: high sphericity, subrounded to rounded, equant, 0.25–0.35 mm. Very minor representation of other phases. The most mature and homogeneous of the four samples.

CLASSIFICATION

Medium-grained quartz arenite, moderately well-sorted, mature to supermature.

PHOTOMICROGRAPHY



Photomicrograph (N+, 2.5× objective) of a grain-supported, moderately well-sorted, medium-grained quartz arenite. Monocrystalline quartz (Qz mono) is overwhelmingly dominant, ranging from subrounded to rounded; polycrystalline quartz (Qz poly), feldspar (F), and lithic fragments (L) are very subordinate. Long contacts predominate, with close grain-to-grain juxtaposition suggesting siliceous cementation via overgrowths; ferruginous cementation is restricted and localized.